Methodology: Implementation Details

This section describes how the formal CIF mechanisms from Part 1 are realized as executable defense algorithms. The implementation follows three design principles: (1) **fidelity to formal specification**—each algorithm directly implements a Part 1 definition or theorem, with explicit cross-references; (2) **composability**—mechanisms operate independently and compose through well-defined interfaces, matching the defense composition algebra; and (3) **configurability**—all thresholds, weights, and operational parameters are externalized to enable deployment-specific tuning (sec:config-params).

> ***Cross-Reference Note**: All algorithms implement formal definitions from Part 1. We cite specific theorems using "(Part 1, Theorem X.Y)" notation to enable traceability from implementation to theoretical foundations.*

The implementation comprises six core algorithms (sec:pseudocode), 27 configuration parameters organized into eight parameter groups (sec:config-params), and 13 packages comprising eight core defense modules totaling approximately 21,000 lines of